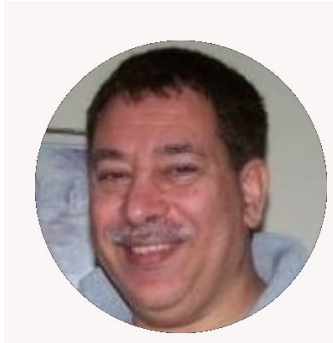


Nicholas (Nico) Michael

| Senior Java Developer Team Lead | Solutions Architect



1. Personal Information

Nationality: South African

Location: Pretoria, Gauteng, South Africa

Languages: English & Afrikaans

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GitHub: <https://github.com/nic0michael>

AI Portfolio video: https://youtu.be/GyjAqyJZw_Y

Professional Summary

Senior Java Developer, Technical Lead and Solutions Architect with **26+ years of Java** development experience and **6 years of architecture experience** across Infrastructure and Solution Architecture. 5 years of Python coding.

Delivered 56+ Spring Boot microservices into production at WesBank

I worked across Banking, Finance, Healthcare, Insurance, Telecommunications, Education, Loyalty and Utility sectors.

Strong practical experience in **Java, Spring Boot, Kubernetes, Docker, Terraform, Azure AKS** and CI/CD.

Development Best Practice Coach specializing in TDD, BDD, Clean Code, DevOps and AI-assisted development.

How do I approach a problem?

I try to understand the problem before jumping to a solution. I ask questions to get the right context, research the problem, consider different solutions and their trade-offs, and then implement and validate the chosen solution.

An example from my time at Cell C:

When I joined Cell C, I found that the Synix reporting system was very slow. The reports took around 30 minutes for all to run because they performed **expensive queries** with many joins against the production database of the cellular system.

I recognized that this was not only a performance problem. Running these queries against the production database could potentially **affect the system supporting customer calls** on the Cellular System.

I redesigned the solution by **introducing a staging database, isolating the reporting workload** from the production system. I also created dedicated reporting tables so that reports did not have to repeatedly execute expensive queries across multiple databases.

The result was immediate: reports that previously took around **30 minutes were reduced to approximately 3 minutes**, while removing the reporting workload from the production system.

There was another architectural problem. **The existing system had been designed for Siemens/Nokia base stations**, but Cell C had started installing Huawei equipment.

Rather than modifying the existing code specifically for Huawei, I changed the design so that the **data-capture components implemented a common interface** and used a Factory Design Pattern. This allowed separate implementations for Siemens/Nokia and Huawei while keeping the rest of the system independent of the underlying technology.

This also meant that future base-station technologies could be introduced without having to redesign the entire system.

This is typical of how I approach problems: I look beyond the immediate issue and try to find a solution that improves the system while also making it more maintainable and adaptable for the future.

Core Competencies

- Software Engineering: Java, **Spring Boot**, **REST** APIs, JPA, Full Stack Development
- Technical Leadership: **Team Leadership**, **Mentoring**, Training, Development Best Practices
- DevOps & Cloud: **Kubernetes**, **Docker**, **Helm**, **Jenkins**, **GitLab CI/CD**, **Terraform**
- Cloud Platforms: **Azure AKS**, **AWS**, **GCP**, **Oracle Cloud**
- Architecture: Solution Architecture, Infrastructure Architecture, Enterprise Architecture
- Cloud-Native: **Kubernetes**, **K3S**, **Docker**, **Helm**, Cloud-Native Architecture
- Software Quality: TDD, BDD, JUnit, Mockito, **SonarQube**, Clean Code
- AI-Assisted Development: Spec Driven Development SDD, Context Driven Development CDD, Prompt Engineering, Multi-Agent Development
- AI Tools: GitHub Copilot, Claude Code, Codex, Cursor
- Process & Architecture: **UML**, **C4**, **TOGAF**, **IBM GS Method**

DevOps & Cloud Engineering Experience

Practical experience across DevOps, cloud infrastructure, automation, containerisation, Infrastructure as Code, CI/CD and Kubernetes.

- 2 years **Azure** experience including Azure AKS Kubernetes and Pipelines.
- 1 year **GCP** experience developing a Notification App using Firebase in GCP.
- 1 year practical **AWS** experience building infrastructure for Udemy training.
- 4 years **Oracle Cloud** provisioning **Rocky Linux** servers
- 8 years **Docker** experience.
- 7 years **Kubernetes** and Cloud-Native platform experience.
- Applied **Terraform** for repeatable Infrastructure as Code deployments.
- Developed PowerShell and Linux shell automation for development, testing, analysis and deployment.
- Worked with **Docker Compose** and **multi-container environments**.
- Practical Kubernetes experience covering K3S, workloads, nodes, namespaces, pods, services and troubleshooting.
- Experience with Proxmox virtualised infrastructure.
- Experience with **SonarQube** for automated code-quality and security analysis.
- Practical troubleshooting of networking, ports, DNS, certificates, containers, virtual machines and deployment failures.
- Applied **DevSecOps** principles within the software delivery lifecycle.

Architecture Experience

Infrastructure Architect: IBM, ESKOM – 3 years

Solution Architect: Cell C, FleetDomain, Vertice – 3 years

Designed and supported enterprise, infrastructure and application architectures across financial services, telecommunications, utilities and healthcare environments.

AI-Assisted Development & Engineering Leadership

Led research, evaluation and adoption of AI-assisted software development tools including GitHub Copilot, Claude Code, OpenAI Codex and Cursor.

- Researched and implemented Spec-Driven Development using Microsoft GitHub Spec Kit.
- Developed corporate AI policy files and coding governance guidelines for AI-generated code.
- Evaluated single-agent and multi-agent development approaches.
- Evaluated competing AI platforms using a common full-stack application and recommended Claude Code and Codex based on practical results.
- Researched local LLM development using Ollama, OpenCode and Google Gemma.
- Created the Iterative Context-Driven Development (ICDD) methodology using AI-assisted brainstorming, requirement refinement and iterative quality reviews.
- Produced technical training content through a DevOps-focused YouTube channel.

Key Achievements

- Delivered **56 Spring Boot microservices** into production at WesBank.
- Designed and implemented a **High Availability K3S Kubernetes cluster** spanning Germany, England and Johannesburg.
- **Developed a Python-based K3S Kubernetes Cluster Generator** to automate Kubernetes cluster configuration and generation.
- Used Helm Charts to automate application deployment and management across Kubernetes.
- Created technical demonstrations **integrating Terraform, Kubernetes, Helm and Proxmox**.
- Introduced TDD, BDD and DevOps best practices across development teams.
- Developed the AHAA Agentic Development Methodology.
- **Lead the integration of the Billing Switch** application into **Azure AKS Kubernetes**.
- Published a Microservices White Paper.
- Published 50 Open Source projects on GitHub, including K3S Auto Cluster Generator and Claude Code Multi Agent With Policies.
- **Provided technical training** in Java, Kubernetes, DevOps and AI-Assisted Development.

Key Projects

WesBank: Credit Scoring System 2018

Role: Senior Java Developer Team Lead

I was hired to deliver a Credit Scoring System, I **analyzed requirements** and designed the solution using UML. Designed a **non-synchronous, parallel-processing architecture** using RabbitMQ across a 4-node cluster, with Oracle for credit-score results and a micro-service for end-user access. Researched and prototyped a Drools rules engine to manage extensive business rules. Led implementation through to Production; The scoring service code remained unchanged for 3 years

IBM: System Architecture for GJMC Metro Council of Johannesburg 2001

Role: System Architect

Took responsibility for the software and networking architecture across the GJMC Metro Council of Johannesburg. The requirement was to develop a complete Systems Architecture covering all customer regions, each with its own existing systems and requirements.

I visited each region to **identify and understand the existing systems** and infrastructure. I then produced a complete set of Architecture artefacts for each region, together with the overall GJMC Systems Architecture and the supporting documentation. This provided a consolidated architectural view of a large, geographically distributed environment

IBM: Architecture deliver for StanLib development project 2002

IBM had been contracted by StanLib to deliver an Insurance System **based on IBM's Insurance UML Model**, with development and implementation to be performed by a subcontractor. My responsibility was to ensure that the proposed system had the required infrastructure architecture to support its development and implementation.

I analyzed the proposed system and created the complete Infrastructure Architecture artefacts and supporting documentation required by the subcontractor. My colleague was responsible for the Solution Architecture; together we delivered the complete solution, including the required infrastructure, to the subcontractor for development and implementation.

Vertice: Dockerizing the Billing Switch Project taking it into the Azure AKS Kubernetes cluster 2025

This was done in 5 phases:

1. **Dockerized the Billing Switch** so new hires could easily install the required applications on their laptops.
2. **Taught Kubernetes to the team** performing the Azure AKS POC, including the Team Leader and DevOps team.

3. **Led the POC** team, creating the Docker container and Kubernetes manifests.
4. **I created the Docker Containers** in our Azure DevOps Repository
5. Worked with DevOps, we created the pipelines and deploy the Billing Switch to the Development environment.
6. **I resolved Issues with the Kubernetes Manifests**, then deployed the Billing Switch to UAT and ultimately to Production.

Employment History

Vertice

Solution Architect | Aug 2025 – Present

Focus: Enterprise Solution Architecture, AI Agentic Systems R&D, AI Architecture, Agent-Oriented Design, Technical Strategy and Innovation.

- Maintained the Architecture Intellectual Capital repository in SharePoint.
- Maintained the Architectural Decision Register (ADR) process and repository.
- Maintained the System Architecture Review (SAR) framework and documentation.
- Researched and evaluated emerging AI-assisted software development tools.
- Researched Spec-Driven Development using GitHub Spec Kit.
- Researched Claude Code, Cursor and Codex.
- Investigated single-agent and multi-agent AI development approaches.
- Developed corporate AI policy frameworks and coding governance guidelines.
- Created and maintained the AHAA methodology, drawing on C4, TOGAF, Zachman and IBM Global Services methodologies.
- Developed Tech-Radar applications providing real-time technology adoption information for management.
- Developed applications providing management with Code Quality, Security Issues and Technical Debt information, project managers with Git repository status, and business analysts with development task status.
- Provided training on Agentic Development tools.

Tech Stack: GitLab, Azure DevOps Repo, Azure AKS Kubernetes, GitHub Copilot, GitHub Spec Kit, Claude Code, Cursor, Codex, VS Code, IntelliJ, Java, Python, Angular, Spring Boot

Vertice

Senior Java Developer | Full Stack Developer | Best Practice Coach | Aug 2024 – Aug 2025

- Developed application code and provided technical training to development teams.
- Led the Dockerization of the Billing Switch.
- Mentored team members in Docker adoption.
- Participated in the Azure AKS Kubernetes POC.
- Guided the team through its first AKS Kubernetes adoption and integration.
- Provided Kubernetes and cloud-native development training.
- Introduced and delivered Helm training to the DevOps team.
- Trained development teams in TDD.
- Introduced and trained teams on Spec-Driven Development.
- Taught Context-Driven Development using Claude Code.
- Introduced advanced brainstorming techniques to improve SDD and CDD outcomes.

Tech Stack: Java, Spring Boot, Docker, Dockge, Kubernetes, K3S, Azure AKS, Helm, GitLab, Azure DevOps

LoyaltyPlus

Senior DevOps Consultant / Senior Developer | Jun 2023 – Jul 2024

- Designed and **implemented Disaster Recovery** and backup strategy for customer WordPress platforms using Ansible, Docker, HAProxy and Semaphore.
- Built and automated **High Availability K3S Kubernetes cluster** across Germany, England and South Africa.
- Developed **infrastructure automation using Ansible, Terraform, Proxmox**, Cockpit and Rocky Linux.
- Developed a distributed messaging platform using Camel Quarkus, Kafka, Angular and REST APIs supporting SMS, Email, WhatsApp, Push Notifications and In-App Notifications.
- Implemented Firebase integration for mobile notification services.
- Designed security monitoring and vulnerability scanning using Metasploit and StarRocks.
- Evaluated and implemented Zero Trust networking using NetBird.

Tech Stack: Java, Spring Boot, **Camel, Quarkus, Kafka, Angular, Docker, K3S Kubernetes, Helm, Ansible, Terraform, Proxmox, HAProxy, Semaphore, NetBird, Metasploit, StarRocks, Google GCP, Firebase**, REST APIs, Rocky Linux Servers.

Advance / MTN

Senior Software Engineer | Kubernetes Specialist | Technical Trainer | Jun 2022 – Jun 2023

- Developed solutions for the MTN USSD platform.
- Designed and implemented **High Availability K3S Kubernetes clusters** on Red Hat infrastructure.
- Created Kubernetes manifests and Helm Charts for Zookeeper, Elasticsearch, Kibana, Prometheus, PostgreSQL and MongoDB.
- Led Kubernetes adoption and deployment practices across development teams.
- Delivered training in TDD, JUnit, Mockito, Kubernetes, Helm and cloud-native deployment practices.

Tech Stack: Java, Spring Boot, **Kubernetes**, K3S, **Helm**, **Docker**, Elasticsearch, Kibana, **Prometheus**, PostgreSQL, MongoDB, Zookeeper, JUnit, Mockito, **Jenkins**, Terraform

DVT

Senior Java Developer Team Lead | Feb 2017 – Jun 2022

Discovery | 2022

- Led Spring Boot Micro-service development initiatives.
- Developed Java training curricula and technical enablement programs.

Tech Stack: Java, Spring Boot Micro-services, Hibernate, GitLab, TDD, Mockito.

Curro Schools | 2021

- Served as Software Development Best Practices Coach and Senior Java Developer.
- Delivered Proofs of Concept covering TDD, BDD, **Docker**, **SonarQube**, Portainer, RabbitMQ, Postman and Micro-service architecture.
- Trained development and QA teams in automated testing and engineering best practices.

Tech Stack: Java, Groovy, Grails, Spring Boot Micro-services, **OAuth**, TDD, **SonarQube**

WesBank | 2018 – 2020

- Delivered 56+ Spring Boot microservices into production.
- Led adoption of TDD, BDD, **SonarQube**, **Bitbucket** and Confluence.
- Designed a KYC Rules Engine using Drools and RabbitMQ.
- Architected integration services for LDAP, App Notifications, CT02, CIS, Actimize and Documentum.
- Produced solution designs, architectural artifacts and technical documentation.
- Received **Kubernetes training** from WesBank
- Completed **AWS** training through Udemy and online practical for Just over 1 Year .

Tech Stack: Java, **Spring Boot**, **SOAP**, **RabbitMQ**, **Docker**, Docker Compose, Portainer, **SonarQube**, **Postman**, JUnit, Mockito, TDD, BDD, **Drools**, **Kubernetes**, Elasticsearch, Logstash, Kibana, LDAP, IBM ODM, **Nexis**.

Sanlam | 2020

- Developed Spring Boot microservices integrated with IBM ODM Business Rules Management Engine.
- Used IBM Rule Designer to develop process flow diagrams.

Tech Stack: Java, **Spring Boot**, JUnit, TDD, **IBM ODM**, GitLab

Momentum | 2017 – 2018

- Was the Full-stack Developer Team Lead
- Responsible for architecture, TDD adoption, code quality.
- Worked with **Docker**, **Docker Swarm** and **Ansible** automation

Tech Stack: Java, **Angular 2**, **Typescript**, **Bootstrap**, **Docker**, Docker Swarm, Postman, JUnit, Mockito, TDD, IBM ODM, Ansible, GitLab

FleetDomain

Solutions Architect | Infrastructure Architect | Senior Java Developer | May 2016 – Jan 2017

- Implemented enterprise security policies, disaster recovery, backup automation and UAT environments.
- Designed High Availability production architecture and hot backup solutions.
- Reverse-engineered and optimised Groovy/Grails enterprise applications.
- Modernised deployment architecture and server provisioning processes.

Tech Stack: Java, Groovy, Grails, Ubuntu, Windows Server, Linux Administration, Disaster Recovery, Security Architecture, Backup Automation

Standard Bank

Senior Java Developer | Jul 2015 – May 2016

- Developed enterprise OTP authentication services for Africa and Offshore Banking platforms.
- Delivered JEE8 interceptor framework and Vasco OTP integration.
- Applied TDD and BDD using JUnit, Mockito, WireMock, JBehave and Serenity.

Tech Stack: Java, JEE8, **SOAP**, **Maven**, JUnit, Mockito, WireMock, JBehave, Serenity BDD, SOAP, Web Services. Centos Linux, Puppet.

Cell C

Java Solution Architect | Infrastructure Architect | Senior Java Developer | Mar 2014 – Jul 2015

- Redesigned enterprise systems for scalability and performance.
- Changed the Architecture of Badly performing system
- Analyzed the Siemens Nokia Synix architecture Produced UML models.
- Optimized the Complete System Design.
- Made substantial improvement to performance

Tech Stack: Java, UML, Enterprise Architect, EJB, Oracle SQL, Centos Linux, Percona MySQL, JUnit, TDD, Javascript.

Signio

Senior Java Developer | Nov 2009 – Feb 2014

- Developed Standard Bank integration gateways and insurance platform integrations.
- Delivered Jasper Reports solutions and Groovy/Grails applications.
- Developed a project for the VAF5 System to send requests to the NATIS gateway
- Delivered new gateways to the Insurance companies

Tech Stack: Java, EJB3, SOAP, Centos Linux, Percona MySQL, Groovy, Grails, Jasper Reports

Paracon

ESKOM

Infrastructure Architect | Senior Java Developer | Apr 2007 – Jul 2009

- Delivered architecture and integration designs for the Online Vending System
- Installed and configured enterprise integration platforms and middleware.
- Developed BPEL Business Processes for SOAP integration connecting SAP using Sun Java CAPS.
- Worked as **Infrastructure Architect** for 2 years.

Tech Stack: Java, **SOAP, BPEL, LDAP**, Sun Java CAPS, Infrastructure Architecture, **Ubuntu** Linux Server, **VmWare ESX Server**

ABSA Bank

Senior Java Developer | Jan 2006 – Mar 2007

- Developed Home Loans portal solutions using JSR168 Portlets and JSF.
- Participated in delivery of the production Home Loans platform.

Tech Stack: Java, JSR168 Portlets, JSF

IBM Global Services

Infrastructure Architect | Java Developer | Feb 1998 – Jun 2002

- Developed components of the National Online Casino Gambling Monitoring System.
- Worked as Infrastructure Architect.
- Delivered **Infrastructure Architecture for StanLib project**
- Created Architectural artifacts for the GJMC project for Johannesburg Metro Council
- I conducted the **CMM Maturity Rating of Rubico** for the World Bank
- Was part of the IBM **GS Method Methodology** training team of IBM

Tech Stack: Java, Java EJB, Java J2EE, RUP, Visual Age for Java, UML Sybase SQL server, IBM DB2, Windows NT Server, O/S2

Education, Professional Training & Qualifications

National Higher Diploma – Production Management

Tshwane University of Technology | Graduation 1989

National Diploma – Electrical Engineering L/C

University of Johannesburg | Graduation 1980

Matric

TED | 1975

Professional Training:

SQL,Java, EJB JEE, UML, C++, Kubernetes, Docker, Jenkins, Terraform, Azure, AWS, Metasploit and Agile Methodologies.

Training providers include **IBM, Damelin and Udemy**

Publications, Open Source & Community Contributions

Published Microservices White Paper on Test Driven Development.

- <https://www.dvt.co.za/white-paper-building-cloud-ready-microservices-using-industry-best-practices>

Published 50 Open Source projects on GitHub.

- Created the K3S Kubernetes Auto Cluster Generator
- The Claude Code Multi Agent Development project With Policies.
- Technical training in Java, Kubernetes, DevOps and AI-Assisted Development.

Published Videos in DevOps YouTube channel

- Videos covering Java, Enterprise Java, **Quarkus**, TDD, **Docker**, **Kubernetes**, Portainer, **Proxmox**, **Terraform**/OpenTofu, **Helm**, **Jenkins**, **Ansible**, AI Agentic Development and N8N.

Professional Memberships

- SAIETE – South African Institute for Technician Engineers
- SAIEE – South African Institute for Electrical Engineers
- JUGSA – Java User Group of South Africa
- GLUG – Gauteng Linux User Group

References

Professional and personal references available upon request.